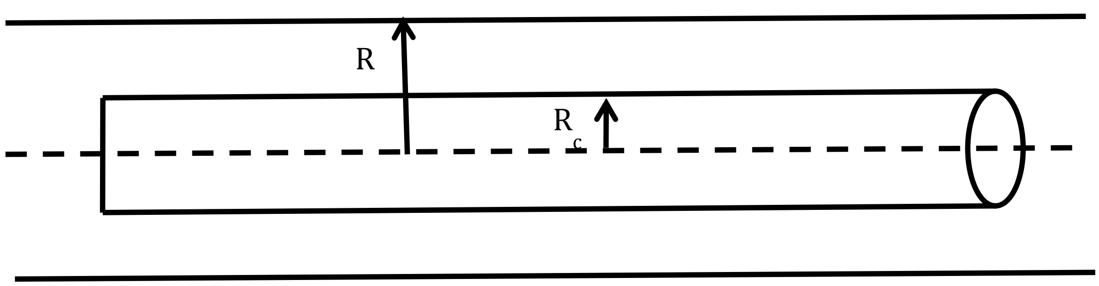
CHE 311 Homework Week 9

1. BSL page 106 3B.4
2. BSL page 106 3B.6
3. Consider a catheter of radius R_c placed in a small artery of radius R as shown in the figure. The catheter moves at a constant speed V. In addition, blood flow through the annular region between R_c and R under a pressure gradient $\Delta p/L$ that only varies in the z direction. We want to determine the effect of the catheter upon the shear stress at $r=R$. Assume steady fully developed flow of a Newtonian fluid.
   (a) State the momentum balance and boundary conditions.
   (b) Solve the momentum balance; substitute Newton's law of viscosity and solve. Apply the boundary conditions to determine the velocity profile.
   (c) Calculate the shear stress.
   (d) Use the following vale to determin the shear stress acting on the blood vessel surface, $r=R$: $R=0.17$ *cm*, $R_c=0.15$ *cm*, $V=10$ *cm/s*, $\mu=0.03$ *g/(cm-s)*, $\Delta p/L=100$ *dyn/cm3*.



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